Forces: independent practice

Mass, weight, contact forces and force diagrams

Name _____ **Class** _____

1 Mass and weight

a) What is mass? Include its unit.

b) What is weight? Include its unit.

2 Contact or non-contact?

Write C for contact or N for non-contact.

Weight:	_____	Friction:	_____	Magnetism:	_____
Support:	_____	Air resistance:	_____	Tension:	_____

3 What affects the force?

a) Which usually produces more friction: rough or smooth surfaces? Why?

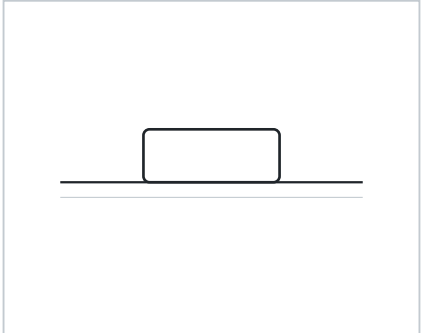
b) At the same speed, which has more air resistance: flat or crumpled paper? Why?

4 Draw the forces

Draw and label force arrows on each object.

Book on desk

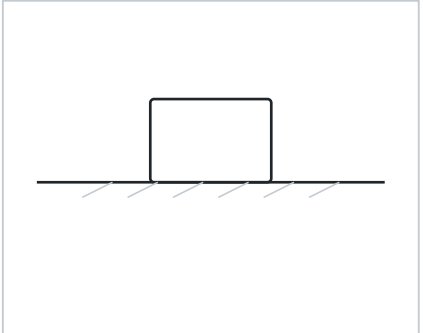
At rest



Balanced? ○ Yes ○ No

Sliding box

Moving right; slowing



Balanced? ○ Yes ○ No

Falling paper

Moving down; speeding up

Balanced? ○ Yes ○ No

Mass, weight and resultant force

Show your working. Use g = 10 N/kg on Earth.

5 From mass to weight

Use W = m × g. Convert grams to kilograms first.

Object	Mass (g)	Mass (kg)	Weight (N)
A	120		
B	350		
C	600		

Show one calculation:

6 Which world?

Gravity (N/kg): Earth 10 Moon 1.6 Mars 3.7 Venus 8.9

Use gravitational field strength = weight ÷ mass.

World	Mass (kg)	Weight (N)	g (N/kg)	Name
A	2	7.4		
B	1	8.9		
C	0.5	0.8		

Why can the same object have different weights on different worlds?

7 Resultant force

Find the resultant, direction and whether the forces balance.

a) 8 N left and 8 N right **R:** _____ **Dir:** _____ **B/U:** _____

b) 12 N right and 5 N left **R:** _____ **Dir:** _____ **B/U:** _____

c) 10 N down and 4 N up **R:** _____ **Dir:** _____ **B/U:** _____